

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the function

$$f(x, y) = (x^3 + y^2)^5$$

a) Find its two first partial derivatives $f_x(x, y)$ and $f_y(x, y)$.

Solution:

$$\begin{aligned}f_x(x, y) &= 5(x^3 + y^2)^4 \cdot (3x^2) = 15x^2(x^3 + y^2)^4 \\f_y(x, y) &= 5(x^3 + y^2)^4 \cdot (2y) = 10y(x^3 + y^2)^4\end{aligned}$$

b) Find its four second partial derivatives $f_{xx}(x, y)$, $f_{yy}(x, y)$, $f_{xy}(x, y)$, and $f_{yx}(x, y)$.

Solution:

$$\begin{aligned}f_{xx}(x, y) &= \frac{\partial f_x}{\partial x} = 30x(x^3 + y^2)^4 + 15x^2 \cdot 4(x^3 + y^2)^3 \cdot (3x^2) \\&= 30x(x^3 + y^2)^4 + 180x^4(x^3 + y^2)^3 \\f_{yy}(x, y) &= \frac{\partial f_y}{\partial y} = 10(x^3 + y^2)^4 + 10y \cdot 4(x^3 + y^2)^3 \cdot (2y) \\&= 10(x^3 + y^2)^4 + 80y^2(x^3 + y^2)^3 \\f_{xy}(x, y) &= \frac{\partial f_x}{\partial y} = 15x^2 \cdot 4(x^3 + y^2)^3 \cdot (2y) \\&= 120x^2y(x^3 + y^2)^3 \\f_{yx}(x, y) &= \frac{\partial f_y}{\partial x} = 10y \cdot 4(x^3 + y^2)^3 \cdot (3x^2) \\&= 120x^2y(x^3 + y^2)^3\end{aligned}$$

Problem 2. (10 points) Given the function $f(x, y) = x^3y + 3xy^2$ and a point $\mathbf{p} = (2, -2)$,

a) Find the gradient vector $\nabla f(\mathbf{p})$.

Solution:

$$f_x(x, y) = 3x^2 + 3y^2, \quad f_y(x, y) = x^3 + 6xy$$

So we have

$$\begin{aligned}\nabla f(\mathbf{p}) &= \langle f_x(\mathbf{p}), f_y(\mathbf{p}) \rangle \\ &= \langle f_x(2, -2), f_y(2, -2) \rangle \\ &= \langle 3 \cdot 2^2 \cdot (-2), 2^3 + 6 \cdot 2 \cdot (-2) \rangle \\ &= \langle -12, -16 \rangle\end{aligned}$$

b) Find the equation of the tangent plane at $\mathbf{p}$.

Solution:

$$f(\mathbf{p}) = 2^3 \cdot (-2) + 3 \cdot 2 \cdot (-2)^2 = 8$$

So the equation of the tangent plane at $\mathbf{p}$ is given by:

$$\begin{aligned}z &= f(\mathbf{p}) + \nabla f(\mathbf{p}) \cdot ((x, y) - \mathbf{p}) \\ &= 8 + \langle -12, -16 \rangle \cdot \langle x - 2, y + 2 \rangle \\ &= 8 - 12x + 24 - 16y - 32 \\ &= -12x - 16y\end{aligned}$$

Or we could rewrite by:

$$12x + 16y + z = 0$$